

AnimalSense: Understanding Beyond-human Sensory Capabilities of Animals via VR Games

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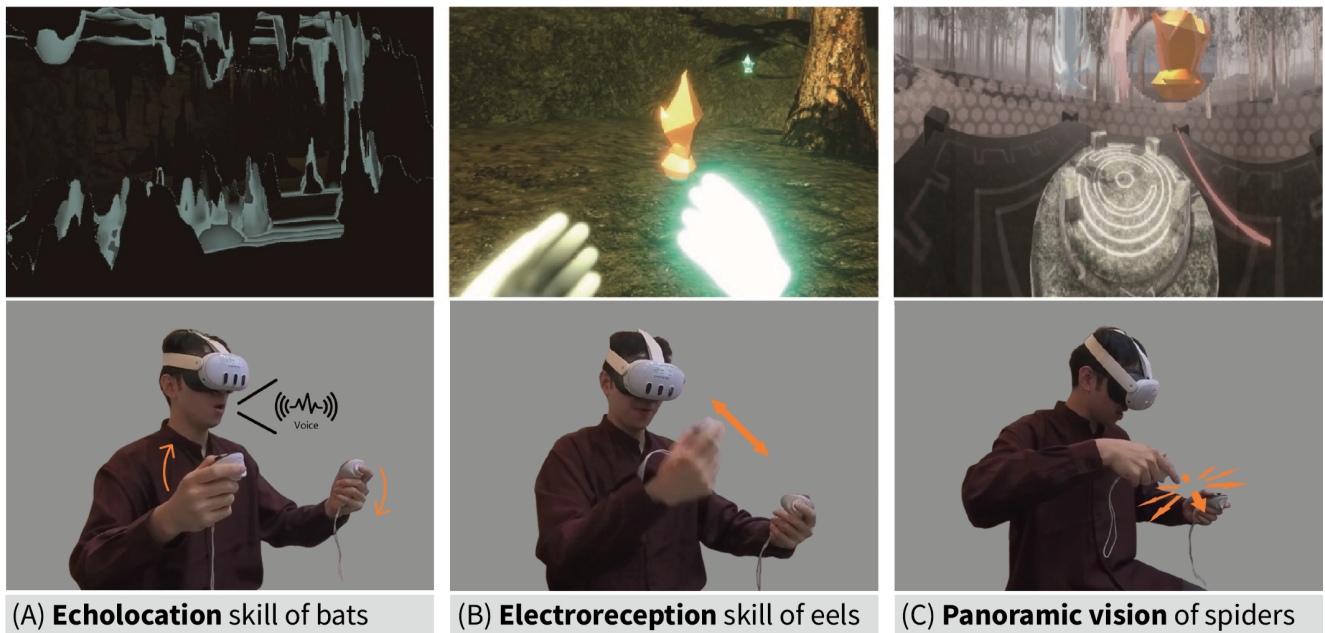


Figure 1: *AnimalSense* is a VR game that enables players to explore and utilize animals' beyond-human sensory capabilities to overcome game challenges, thereby enhancing players' understanding of animals' unique senses. This paper presents three example section designs: (A) bats' *echolocation* to avoid obstacles; (B) eels' *electroreception* to perceive hidden objects; and (C) spiders' *panoramic vision* to see 360-degree environment without the need to turn the head.

ABSTRACT

Animals have sensory capabilities far beyond those of humans, including the ability to detect ultrasound, UV light, electric fields, magnetic fields, and to have panoramic vision. *AnimalSense* is a VR game crafted to immerse players in the extraordinary sensory

worlds of animals. The game's design leverages the concepts of *sensory substitution*, *sensory remapping*, and *active learning*, allowing users to explore and utilize these beyond-human sensory capabilities to overcome game challenges, thereby enhancing players' understanding of animals' unique senses. This paper presents three such game section designs: 1) *echolocation* in bats, 2) *electroreception* in eels, and 3) *panoramic vision* in spiders, showcasing how these are integrated into the gameplay.

CCS CONCEPTS

• Human-centered computing → Virtual reality; • Applied computing → Computer games; • Software and its engineering → Interactive games.

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KEYWORDS

Virtual Reality, Sensory capabilities, Animals, Animal Comprehension, Perceptual ranges

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1 INTRODUCTION

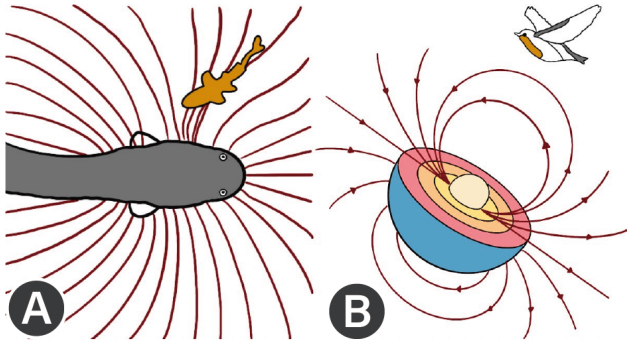


Figure 2: Examples of sensory modalities not present in humans: (A) eel could sense the electric field [8]; (B) migratory bird could sense the magnetic field [18].

There are over 2.16 million living animal species described in scientific literature, including 1.05 million insects and 74,420 vertebrates [28]. They have sensory capabilities far beyond those of humans, both in terms of sensory modalities and perceptual ranges.

Regarding sensory modalities absent in humans, this includes the ability to detect electric and magnetic fields, as shown in Figure 2. For instance, eels utilize electric fields to both generate electrical power and sense disturbances, enabling them to locate prey hidden behind obstacles [9]; migratory birds navigate across thousands of kilometers without losing their way by sensing the Earth's geomagnetic fields [10].

Regarding perceptual ranges beyond human perception, this includes animals' vision, auditory, and olfactory systems that can sense significantly wider and farther than those of humans, as depicted in Figure 3. For instance, fish have a nearly 360-degree field of view (FOV), while spiders possess a 330-degree FOV, both of which far exceed the human horizontal FOV of 200 degrees. [34]; humans can typically hear sound frequencies ranging from 20 to 20,000 Hz, while bats and dogs can perceive frequencies higher than 20,000 Hz; bees perceive visible wavelengths ranging from 300 to 600 nanometers, allowing them to see ultraviolet light, unlike humans whose vision spans from 380 to 750 nanometers.

Traditional learning methods, such as paper and video-based, make it challenging for humans to comprehend beyond-human animals' capabilities. Virtual Reality (VR) could immerse users in a completely different world beyond reality [29], offering the potential to offer novel experiences into how animals perceive the world.

Previous studies have explored animals' visual perspective [20] and unique abilities of body structures [22, 23] in VR. However, most of them are limited to human-centric vision and lack practical use of beyond-human capabilities. Therefore, **we propose *AnimalSense*, a virtual reality game designed to immerse players in the sensory capabilities beyond human perceptual ranges. By challenging players to experience these capabilities to accomplish tasks, the game extends our understanding of animals.**

We investigated various sensory capabilities of animals. For sensory modalities not found in humans, we focused on eels, while for perceptual ranges exceeding human capabilities, we studied bats and jumping spiders. Our aim is to explore how to adapt the capabilities of these three animals to fit within the human perceptual range and enable users to utilize them within the game. Considering that current VR headsets provide visual, auditory effects, and haptic feedback from controllers, we devised designs utilizing each unique skill of the three animals:

- **Bat:** Bats could detect distance or invisible objects by interpreting reflected sound waves, such ability is named *echolocation* [2]. *AnimalSense* allows users to emit sounds and visualize the resulting echo in a dark cave, mimicking how bats use this skill to perceive the world.
- **Eel:** Eels rely on *electroreception* to perceive their environment by generating electric fields [9]. In *AnimalSense*, we replicate this ability using the X-ray vision effect [3, 36] to simulate the interpretation of disturbances in the electric field. Users can utilize controllers to charge and release an electric field, thus experiencing X-ray vision akin to the capabilities of eels.
- **Jumping Spider:** Jumping spiders exhibit an impressive horizontal field of view of approximately 330° [30]. *AnimalSense* uses a concept similar to eyefish lens [4, 37], where we implement *panoramic Vision* to compress a 360-degree field of view into the field of view of a headset, enabling users to grasp how spiders perceive their surroundings without needing to move their heads.

To help players understand the sensory capabilities of three animals, we adopted two learning and training concepts:

- **Sensory Substitution and Remapping:** *AnimalSense* offers visualizations of sensory information beyond human perception since there exist substantial differences in perceptual ranges between humans and animals. This includes visualizing sound waves and employing X-ray vision, as well as incorporating spatial auditory effect in the *panoramic vision*. Besides, *AnimalSense* remapped the 360-degree field of view to a headset field of view. The remapping concept is similar to converting UV images to visible light spectrum [35].
- **Active Learning:** In *AnimalSense*, users engage with the animals' capabilities to tackle various tasks, enhancing their comprehension of animal behavior and characteristics [5]. Leveraging a first-person view in VR heightens immersion and proves advantageous for educational purposes, as it allows users to directly experience the world from the perspective of the animal, facilitating a deeper understanding of their behaviors and adaptations [11, 13].

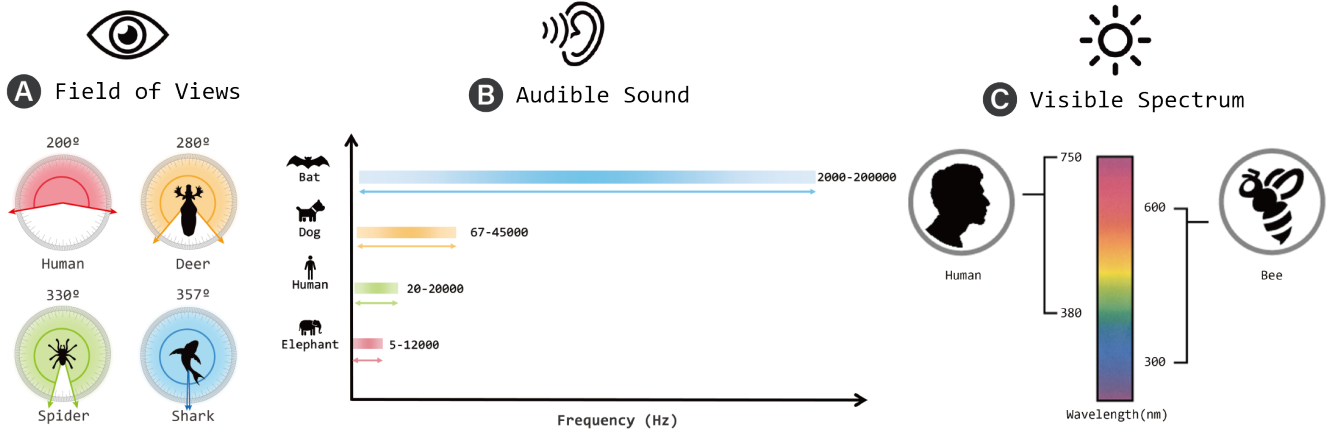


Figure 3: Comparison of perceptual ranges of human vs. animals in terms of: (A) field of view [7, 25, 30, 34], (B) audible sound [6], and (C) visible spectrum [27].

In conclusion, this work facilitates players' exploration of the diverse and unique sensory capabilities possessed by animals beyond human perception. Through visualization, auditory effects, and practical application, users gain an immersive understanding of how animals perceive and interact with their environment.

2 GAME CONCEPT AND DESIGN

AnimalSense was designed to facilitate a novel sensory experience for users, drawing inspiration from the sensory capabilities of various animals. This includes 1) the *echolocation* of bats, 2) the *electroreception* of eels, and 3) the *panoramic Vision* of spiders, as shown in Figure 4. **We have transformed these features into visually and audibly accessible forms that align with the way humans perceive, a process also known as Sensory Substitution and Remapping**[12], and have made them applicable to standard devices without the need for additional equipment, enhancing the feasibility and reach of educational initiatives.

2.1 Bat

Bats predominantly rely on *echolocation* rather than vision for environmental detection. They emit high-frequency sound waves from their mouths and noses, and by interpreting the echoes as these waves bounce off objects, they discern the spatial layout of their surroundings [2].

- **Echolocation → Visualizing Sound Waves in Darkness**

We have set the scene in a dark cavern, mirroring the poor visual acuity of bats. As players make sound, they can observe luminescent sound waves that propagate from near to far, delineating the terrain's topography. This mechanism enables environmental perception through sound, allowing users to navigate and avoid rocks within the scene. Previous works used background or real-world music to create sound waves depicting the contours of the environment, aiming to enhance the atmosphere of horror and suspense [15, 21, 31, 33]; while *AnimalSense* uses similar visualization to mimic how bats use this skill to perceive the world. Figure 5(B) demonstrates the visualization of sound waves overlaid on the same

view in Figure 5(A), revealing how luminescent waves aid in identifying obstacles.

- **Gliding Control**

Users can navigate left or right by controlling the height difference between two controllers. For example, when the left controller is elevated above the right one by a certain threshold, the user gradually glides to the right.

2.2 Eel

Eels possess the remarkable capability to perceive their environment through the generation of electric fields, facilitated by the release of electric currents. By interpreting these electric fields, they can effectively detect and navigate their environment.

- **Electroreception → X-Ray Vision Ability**

The primary feature of the electric eel's *ensory capabilities* is to locate various materials by emitting electric fields through low-voltage discharges. We have adapted this into an X-ray vision ability [3, 36] in our game design. The user initiates the process by charging the electric power by shaking the controller. Once charged, they can release the electric power, granting them the ability to temporarily visualize crystals hidden within the terrain. In Figure 5(C), only visible objects are discernible to the human eye, while in Figure 5(D), X-Ray vision, mimicking the capability of an eel, allows the visualization of objects concealed behind the mountain.

- **Charging & Releasing Electric Field**

When thinking about electric power, one naturally associates it with kinetic energy. Thus, we designed that users accumulate electric power through the kinetic action of shaking controllers. Once a predetermined threshold of accumulated electricity is reached, users can then release electric fields, resulting in the activation of X-ray vision. To enhance the entertainment, we've added one additional function to the electric power, allowing it to serve as a means of distinguishing genuine from fake crystals. Users can release electric power near a crystal, and if it's fake, the crystal will shatter.

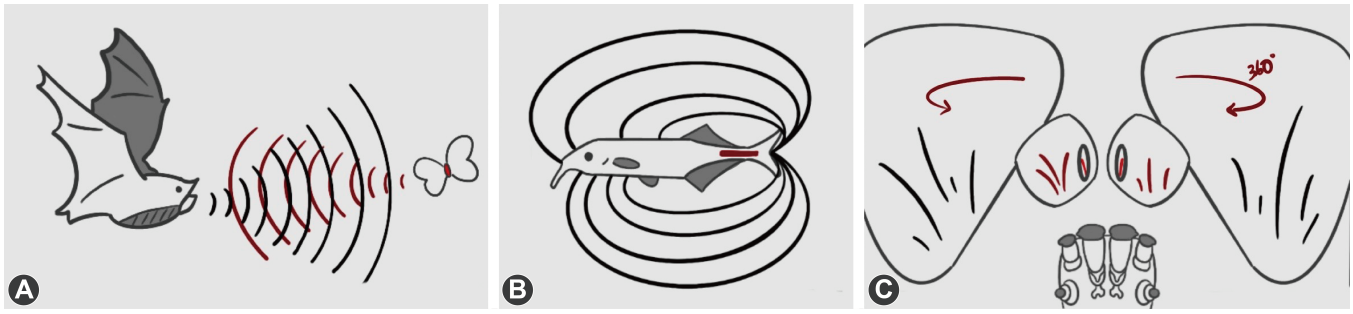


Figure 4: Different sensory capabilities of selected 3 animals: (A) bats using *echolocation* to detect the prey [26]; (B) eels using *electroreception* to perceive the environment [16]; (C) jumping spiders using near 360-degree *panoramic Vision* to detect moving objects [32]

2.3 Spider

Certain spiders in the *Salticidae* family possess visual systems comprising principal eyes that provide focused and detailed vision with high resolution to complete main tasks [17]. Additionally, they possess secondary eyes contributing to nearly 360-degree *panoramic vision*, enabling the detection of motion to detect risks immediately, albeit at lower resolution. *AnimalSense* has designed corresponding sections for this kind of principal-eye and secondary-eye experiences.

- **Panoramic Vision → Compressed 360° scene**

To present the aforementioned visual effects to human users, we have compressed the 360-degree panoramic field of view into the scope of VR headsets. Given that humans possess only a 200-degree field of view, it's challenging for us to understand the concept of a jumping spider's nearly 360-degree vision. Thus, in our game design, we use 6 cameras in Unity and map those views onto a plane to build a 360-degree panorama scene, which uses real-time rendering techniques to allow players to explore what it would be like to have the *Panoramic Vision*. To mimic how jumping spiders utilize their wide field of view to detect threats, we incorporate fireballs with spatial audio attacking the player from eight directions. With the incorporation of *panoramic vision* simulating the spider's wide field of view and auditory effects, users can effectively observe incoming fireball attacks from all directions without the necessity to turn their heads. In Figure 5(E), the limited field of view (FOV) of humans prevents them from seeing the entire environment, whereas in Figure 5(F), *panoramic vision* enables the identification of risks from a 360-degree perspective.

- **High-resolution vision for main tasks**

For a spider, only the pair of principal eyes in the front provide the clearest view. When a spider is eating its prey, the principal eyes focus on it, while the other secondary eyes detect risks from other positions with lower resolution. *AnimalSense* incorporates a small central area with higher color saturation, clarity, and a long-focus lens in the middle of users' vision to bionically replicate principal eyes. The players use this high-resolution central vision to accomplish the main tasks, which involve using the controller to draw a

trajectory on a magic board and launching iceballs toward the corresponding direction from which the fireballs attack. In short, players utilize the high-resolution central vision to launch the iceball on the magic board while simultaneously employing *panoramic vision* to observe the origin of the fireball.

2.4 Game Flow

AnimalSense is structured into three sections, each prefaced with an instructional tutorial of approximately 30 seconds, followed by a practical use segment ranging from one to three minutes. The game is set against a forest backdrop, wherein players assume the roles of witches or wizards tasked with the guardianship of the forest. In the beginning, the player receives a letter warning that the forest is at risk. To protect the forest, the player must collect crystals and place them on the magic board to create a protective barrier. To collect the crystals, the players borrow the sensory capabilities of a specific animal. In the Bat section, players navigate a dark cavern by employing *echolocation* and traverse this environment to acquire a bat-like crystal. In the Eel section, players search for eel-like crystals in the swamp. However, the crystals are hidden everywhere, and fake crystals are mixed in. Therefore, players utilize *electroreception* to locate hidden crystals and use electric power to test the truth. In the final Spider section, crystals are placed on the magic board. Before the crystals create the protective barrier, players need to protect the magic board from fireball attacks. Players use *panoramic vision* to detect the origins of fireballs and central vision to draw trajectories for launching iceballs in the corresponding direction. Once the task is completed, the crystals form the protective barrier, ensuring that the forest is no longer exposed to danger, thus positioning the player as the savior of nature in this game.

2.5 Implementation

We used Unity and Oculus Quest 2 and 3 headsets to develop our game. Most of the Outlook resources are from the Unity Asset Store with proper licenses. In the Bat section, we utilize the Unity audio tool to measure the loudness of the audio and the Shader Graph to change color in response to depth of contact. In the Eel section, we record the movement speed of the right controller. We

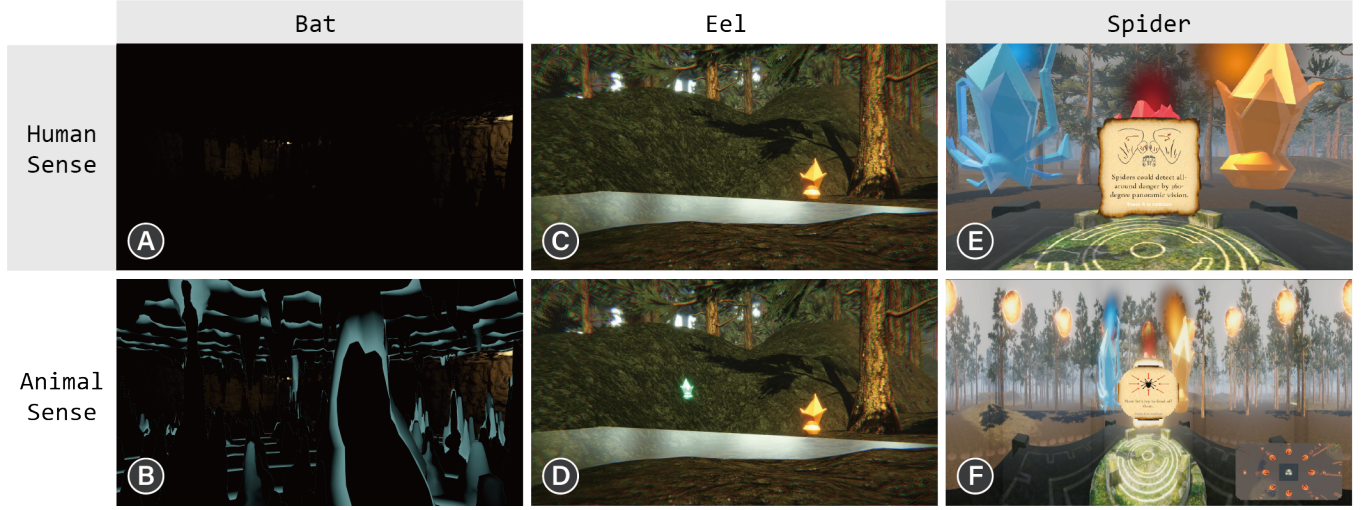


Figure 5: The game-view of *AnimalSense*. (A)(C)(E) represent the vision using pure human sensory capabilities, while (B)(D)(F) represent the vision using the sensory capabilities of animals. For the bat section, users experience the dark caves (A), where after making sound, luminescent sound waves aid in visualizing obstacles (B). For the eel section, users initially see only one crystal in (C), but after charging and releasing electric power, additional hidden crystals become visible (D). For the spider section, users are unable to see the entire environment with the limited human field of view (E). However, after utilizing the *panoramic vision* feature akin to a spider’s, users gain the ability to perceive fireballs from all 360 degrees (F). During tutorial mode, we introduce a top-view window positioned at the lower-right corner, allowing users to familiarize themselves with launching iceballs in each direction.

use this value as a condition to accumulate electric power when it exceeds a predefined threshold. As for the emulation of the spider’s 360-degree field of view, we modified the asset [14] and used multiple strategically positioned cameras. Each camera underwent individualized configuration adjustments, including color, aperture, and blur settings, aimed at providing players with a precise and immersive visual experience of the spider.

3 CRITICAL REFLECTION

AnimalSense allows users to experience various sensory capabilities of animals. Given that these abilities exceed human perceptual ranges, we employ the principles of *sensory substitution* and *remapping*, offering potential applications across various capabilities. In this paper, we explore three examples that illustrate how to utilize a sensory modality not present in humans and two capabilities that exceed human perceptual ranges in game design. These examples serve to investigate and understand how to integrate animal capabilities into game design, providing humans with a deeper insight into animals. Future research can further delve into the design of various animal abilities.

Our research indicates that users can readily experience *echolocation* through the visualization of sound waves, and *electroreception* through X-ray vision, which resembles interpreting disturbances in electric fields. However, during the design process of the spider section, we discovered that *panoramic vision* becomes challenging for players to utilize effectively when fireballs come from every direction. Consequently, we conducted tests to determine the optimal number of directions that strike a balance between playability and challenge. As a result, we made the fireballs come from eight

directions and added spatial audio effects on the fireballs to enhance user understanding. Additionally, in tutorial mode, users can familiarize themselves with the *panoramic vision* by utilizing a top-view window positioned at the lower right corner, as depicted in Figure 5(F). Despite simplified paths and additional visual and auditory cues, users require time to adapt to distinct visual capabilities. Future enhancements aimed at shortening the learning curve may involve integrating haptic feedback to enhance users’ understanding of these unique visual perceptions. For example, implementing vibrotactile feedback designs around the head [38].

In our work, we utilize VR headsets to deliver the gaming experience. It’s noteworthy that VR headsets offer a narrower field of view compared to human vision and lack built-in haptic feedback devices. While VR experience with multisensory stimulation holds potential for educational purposes and for exploring more sensory modalities and perceptual ranges beyond human perception [1, 19, 24], it remains unclear how to effectively integrate *sensory substitution* or *sensory remapping* concepts for perception like haptic or olfactory senses. Moreover, the incorporation of additional senses such as olfaction may require extra devices and could potentially compromise accessibility. Therefore, further research and careful consideration of trade-offs are necessary to explore multisensory approaches aimed at investigating beyond-human sensory capabilities.

4 CONCLUSION

Our work offers an immersive VR game, allowing users to comprehend how animals utilize various sensory capabilities to perceive their environment. Using the concept of *sensory substitution*, *sensory remapping*, and *active learning*, *AnimalSense* enables users to

employ these skills to tackle tasks that are beyond human capabilities and enhance the understanding of animals. We believe that a multisensory VR experience holds the potential to explore sensory modalities that are absent in humans and extend beyond human perceptual ranges. However, further research and careful consideration of trade-offs are necessary to fully understand the implications and possibilities of such exploration.

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